

Measuring Faithfulness in LLM-Generated SHAP Narratives: A Case Study on Adult Income

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Abstract

LLM-generated narratives grounded in SHAP attribution tables are increasingly used to communicate model predictions to the people those predictions affect, yet whether these narratives faithfully represent the numeric source they paraphrase has not been systematically measured. This paper presents a case study on the Adult Income benchmark using Llama-3-70B-Instruct, implementing a four-class taxonomy of faithfulness failure (sign inversion, rank swap, omission, and feature fabrication) alongside systematic prevalence estimates with confidence intervals and prompt strategy treated as a controlled experimental variable. Three-quarters of valid narratives exhibit at least one hallucination under zero-shot prompting. Chain-of-thought prompting substantially reduces overall hallucination rate and nearly eliminates sign inversion, but rank swap persists as the dominant failure mode under both conditions and omission rates are statistically unchanged. Feature fabrication is absent under both conditions, a finding attributed to the explicit feature list in the prompt functioning as a vocabulary constraint. Generalisation to other LLMs, datasets, and underlying classifiers requires further work. Within this case study, the findings indicate that chain-of-thought prompting can address locally verifiable directional errors but leaves structural feature-selection failures largely intact.

1. Introduction

Shapley Additive Explanations (SHAP) assign each input feature a signed numeric contribution to a model's prediction (Lundberg & Lee, 2017), and large language models are now used to convert these tables into natural-language narratives for the people the prediction affects (Meske et al., 2025). The translation step is consequential. A SHAP table is a verifiable artefact. The narrative that paraphrases it is not, and whether the narrative preserves what the table says has not been systematically measured.

The narratives work, with previous research showing that users who received LLM-generated SHAP narratives for classification predictions answered comprehension questions significantly more accurately than users who received the attribution table alone, and more than 90% of participants rated the narratives as convincing (Martens et al., 2024). A narrative that reads as credible and improves comprehension is one users are likely to trust. If that narrative misrepresents the underlying attribution values, the persuasiveness becomes a hazard rather than a benefit. The more credible the narrative form, the more consequential a faithfulness failure becomes.

The paper contributes (a) a four-class taxonomy of faithfulness failure for narrative XAI, (b) a systematic prevalence estimates with confidence intervals across a full evaluation set, (c) prompt strategy treated as a controlled experimental variable rather than a design choice, and (d) an extraction-and-comparison pipeline. Findings are based on a single generator (Llama-3-70B-Instruct) and a single benchmark (Adult Income) and should be read as a case study; generalisation to other LLMs, datasets, and underlying classifiers requires further work. Section 2 reviews related work. Section 3 describes the experimental design, including data, models and prompts. Section 4 presents the results and Sections 5 discusses and concludes.

2. Related Work

SHAP (Lundberg & Lee, 2017) is a widely adopted post-hoc attribution method for tabular predictions, commonly applied in credit scoring, clinical decision support, and algorithmic auditing. While stability issues under feature correlation are documented in the literature (Aas et al., 2020), they are not the focus of this paper. Instead, the current work treats SHAP as a ground-truth baseline for LLM generation rather than analyzing its standalone explainability power. However, SHAP's standard output consists of signed, technically dense numeric scores. As Kim et al. (2024) observe in their systematic review of human-centered explainable AI, local feature importance metrics do not always align with user comprehension. In intricate tasks like financial fraud detection, domain experts find individual feature attributions insufficient because isolating variables disrupts the cohesive narrative

needed to understand a holistic fraud scenario. Consequently, natural language generation (NLG) offers a promising communication layer to translate complex attribution outputs into clear, narrative prose.

However, LLMs are known to hallucinate when generating output. Hallucinations in LLMs broadly divide into two categories. Factual hallucination describes outputs that contradict world knowledge. Faithfulness hallucination describes outputs that contradict a provided grounding source (Ji et al., 2023; Maynez et al., 2020). This paper concerns the latter exclusively. While existing evaluation frameworks have primarily assessed faithfulness in text-to-text settings. Most often, they rely on weakly-supervised classification (Kryściński et al., 2019), sentence-level Natural Language Inference (Laban et al., 2021), or optimized question-answering pipelines (Fabbri et al., 2022) to detect inconsistencies. In contrast, we evaluate adherence to structured inputs. The SHAP table is the complete grounding source, every claim the narrative makes is checkable against it, and world knowledge is irrelevant.

A plausible explanation and a faithful one are not the same thing, and in high-stakes settings the divergence has direct consequences. While plausibility dictates how convincing and accessible an explanation appears to a human user, faithfulness requires that the explanation accurately represent the true causal decision-making process of the model (Jacovi & Goldberg, 2020). In high-stakes domains, optimizing purely for user-friendly, highly coherent explanations often comes at the cost of actual structural truth (Agarwal, Tanneru, et al., 2024). When an unfaithful explanation masks a model's flawed internal reasoning with a highly plausible narrative, it creates a 'worst-case scenario' of unwarranted over-trust, potentially leading to catastrophic failures in deployment (Jacovi & Goldberg, 2020).

Martens et al. (2024) introduced XAIstories, LLM-generated narratives grounded in SHAP values or counterfactuals for tabular classification predictions. In a user study, over 90% of participants rated the narratives as convincing, and users who received narratives answered comprehension questions significantly more accurately than users who received raw SHAP tables. The study did not measure whether the narratives accurately represented the SHAP values they were generated from. Ichmoukhamedov et al. (2024) proposed automated metrics for XAI narrative evaluation using a two-LLM pipeline, where a generation model produces a narrative from a SHAP table and an extraction model parses it into structured feature claims covering rank, sign, and value. Extraction accuracy is validated through manipulated and human-written narratives, though the authors acknowledge these estimates require larger-scale confirmation. A notable limitation is that the faithfulness metrics conflate genuine errors with linguistic ambiguity, as cases where rank or sign cannot be clearly inferred from the narrative text are treated identically to outright mistakes. Manual

analysis on a subsample identified hallucination cases but rates were not reported systematically, and prompt strategy was not treated as an experimental variable.

Two recent studies measure SHAP-adjacent faithfulness in LLM-generated explanations.. AlMarri et al. (2025) evaluated four open-source LLMs as zero-shot classifiers on financial tabular data, computed SHAP values over the LLMs' own predictions, and compared those SHAP values to the LLMs' self-reported feature impacts. Sign agreement between self-explanation and SHAP reached only 50 to 60% even for the most important features. Kendall's alignment with LightGBM SHAP was near zero. Matton et al. (2025) defined causal concept faithfulness as the alignment between the concepts an LLM explanation implies are influential and those that causally affect model outputs, measured via counterfactual inputs and a Bayesian hierarchical model. GPT-3.5 explanations systematically over-cited behaviour-related concepts and omitted identity-related ones regardless of their causal relevance. In both studies, the LLM is the model being explained and also the source of the explanation. This paper asks whether LLMs faithfully relay a numeric grounding source produced by a separate classical model. The ground truth is fully external, and every directional claim is verifiable without causal inference.

Whether chain-of-thought prompting reduces faithfulness failures in SHAP narratives is an open empirical question. Wei et al. (2023) showed that chain-of-thought improves structured reasoning on multi-step tasks. Turpin et al. (2023) showed it can simultaneously surface reasoning that is coherent but unfaithful to the model's actual computation. In SHAP narration, neither outcome can be assumed. We test chain-of-thought as a treatment variable, not as a solution.

3. Experimental Design

Our experimental framework evaluates how prompt engineering affects the faithfulness of LLM explanations. The design crosses a single narrative generation model with two distinct prompting strategies across 150 test instances, yielding a total of 300 generated narratives. Because each instance is processed under both prompting conditions, the prompt strategy functions as a within-instance manipulation. Specifically, each Adult Income instance contributes exactly two narratives (one per prompting strategy) derived from the identical underlying SHAP attribution table. This paired experimental structure establishes the statistical basis for the McNemar tests reported in the results.

3.1. Dataset

We conduct the evaluation on the Adult Income dataset (Barry Becker, 1996), a widely utilized binary classification benchmark that predicts whether an individual's annual income exceeds \$50,000 based on census and demographic records. Specifically, we employ the preprocessed version provided

by the OpenXAI benchmark library (Agarwal, Ley, et al., 2024). We note that recent work suggests retiring the UCI Adult dataset due to concerns surrounding its age, lack of external validity, and flawed feature encodings in algorithmic fairness tasks (Ding et al., 2022). However, the current experimental design treats the underlying model and its data strictly as a fixed, quantitative grounding source to measure LLM linguistic fidelity, rather than evaluating the standalone predictive power or fairness of the classifier. Because the Adult dataset features a standardized schema with easily interpretable demographic features, it remains a highly effective environment for isolating how prompt strategies impact set-membership and directional alignment in natural language generation.

The version selected for the experiment features a standardized, pre-split feature set, a pretrained logistic regression classifier, and precalculated per-instance SHAP attribution values. Following the OpenXAI preprocessing pipeline, each instance is characterized by eleven features:

- **Continuous Features (4):** age, capital_gain, capital_loss, hours_per_week
- **Binary Indicators (7):** sex_Male, workclass_Private, marital_status_Non_Married, occupation_Other, relationship_Non_Husband, race_White, and native_country_US

Methodologically, the survey-weight column `fnlwgt` (final weight) is excluded from the input feature space. This exclusion is standard practice in machine learning benchmarks using this dataset (Kamiran & Calders, 2012). The final weight is a demographic sampling weight calculated by the Census Bureau to represent the population density of a given demographic slice, meaning it does not reflect an intrinsic, predictive characteristic of the individual. We draw 150 test instances as a simple random sample without replacement from the full OpenXAI processed test split (9045 rows). The sample is sufficient to detect the large differences observed across the two strategies - zero-shot (“direct”) prompt and chain-of-thought.

Within the prompt template, features are presented to the LLM in descending order of their signed SHAP values (from most positive to most negative). Binary indicators are displayed as integer values followed by a human-readable categorical label (e.g., `sex_Male: 1 [Male]`), mapped using a fixed, pre-defined metadata registry to ensure syntactic consistency.

3.2. Large Language Model Selection

We partition the pipeline into two distinct phases using separate model architectures: a high-capacity generator to compose the narrative explanations, and a smaller, highly restricted extractor to evaluate the output’s factual alignment with the underlying data.

- **Narrative generation:** We use meta-llama/Meta-Llama-3-70B-Instruct (AI@Meta, 2024). Given its 70-billion-parameter size, this model cannot be reliably hosted on the Hugging Face serverless tier for all concurrent users; requests are therefore routed to third-party high-throughput inference endpoints. We set temperature at 0 for the generation to minimize stochastic variance and maximize deterministic output consistency.
- **Extraction:** We utilize mistralai/Mistral-7B-Instruct-v0.3 (Jiang et al., 2023) with temperature at 0. Utilizing an extraction model from a separate model family (Mistral rather than LLaMA) guarantees cross-family independence, mitigating any shared biases or stylistic loops inherent to a single developer’s model family. Furthermore, because the information extraction task is highly schema-constrained, it does not require frontier-tier reasoning capabilities, making a 7B model both sufficient and highly cost-effective compared to closed-source commercial APIs.

Table 1. Large Language Model Specifications for Experimental Pipeline

Model	Role	Provider	Parameters	Temperature
Llama 3 70B Instruct	Generation	Hugging Face (via Novita)	70B	0.0
Mistral 7B Instruct v0.3	Extraction	Hugging Face Inference Endpoint	7B	0.0

3.3. Prompt strategies

To isolate the effect of reasoning scaffolding on narrative reliability, we cross the generation model with two distinct prompting strategies.

- **Direct Prompting (Zero-Shot):** The first prompt strategy closely follows the narrative template introduced by Martens et al. (Martens et al., 2024), a zero-shot approach. The study is directly concerned with SHAP-to-narrative generation and it has been applied to real user populations, and replicating its prompt design allows the faithfulness findings to be mapped back to a method that already has empirical evidence of adoption. We apply minor syntactic adaptations to accommodate the specific Adult Income feature schema, keeping the core task

instructions, structural constraints, and linguistic register unchanged. The generation model is provided with:

- The binary prediction task description.
 - The model's predicted class and predicted probability.
 - A SHAP table sorted in descending order of signed value.
 - A direct instruction to draft a fluent, cohesive story highlighting the most influential positive and negative features.
- **Chain-of-thought:** The second prompting strategy provides the identical task context, prediction details, and sorted SHAP table, but introduces structural reasoning constraints prior to prose generation. The explicit intermediate steps (sorting, grouping, and sign verification) are designed to reduce set-membership and directional inversion errors by forcing the model's self-attention layers to process the quantitative SHAP tables before generating natural language prose. Before drafting the story, the model is strictly required to execute four sequential steps:
 - Rank the top five features by absolute impact, $|SHAP|$.
 - List all features classified as having negligible impact.
 - Explicitly verify the identity and sign of the single most influential feature.
 - Compose the final explanatory narrative.

3.4. Hallucination Types

The four failure types map onto a structural distinction. Sign inversion and rank swap are attribution errors, in which the narrative misrepresents a feature present in the source by assigning it the wrong direction or by displacing it from the set of most important features. Feature fabrication is an invention error, in which the narrative mentions content with no counterpart in the grounding source. Omission is a coverage error, in which the narrative fails to include source content that is important. This structure divides the types by whether the error is one of commission (attributing incorrectly or inventing) or of non-coverage (failing to relay what the source shows). Types are mutually non-exclusive and a single narrative may exhibit more than one. Relative effect size is not treated as a separate type. Detecting whether a narrative's implied magnitude is proportional to the

SHAP value would require a graded scale rather than a binary flag and falls outside the scope of this paper.

To evaluate whether a generated narrative emphasizes the same key drivers as the underlying explainer, we define the rank swap metric as a set-membership consistency check over the top- k features (default $k = 3$). Rather than enforcing a strict ordinal comparison, we compare the set of k features with the highest extracted importance in the prose against the k features with the largest absolute SHAP values. A rank swap is flagged if and only if these two sets contain different elements, ignoring their internal order. This set-membership approach is intended to partially address the ambiguity concern raised by Ichmoukhamedov et al. (2024), where rank errors and linguistically ambiguous formulations could not be distinguished. This metric works in tandem with an Omission metric. While Omission penalizes models for ignoring critical features entirely, rank swap flags cases where the narrative actively attributes importance to the wrong features. A set-based rule ensures we isolate feature-selection fidelity without double-counting minor ordering shifts.

Table 2. Taxonomy and Detection Criteria for Faithfulness Failures

Type	Definition	Detection summary
Sign inversion	Narrative states the wrong directional effect of a feature (positive vs. negative class)	Extracted sign vs. sign of SHAP value
Rank swap	The set of top- k extracted features (by narrative rank) does not match the top- k SHAP features	SHAP
Feature fabrication	Narrative mentions a feature not present in the dataset’s feature set	Non-empty <code>unknown_features</code> list in extraction JSON
Omission	At least one top- k SHAP feature ($k = 3$ by default) is absent from the narrative	Top- k SHAP features absent from extraction

3.5. Detection procedure

For each narrative, the extraction model receives the narrative text, the full feature list for the dataset, and the prediction task description. It returns a JSON object in which each feature mentioned in the narrative is recorded with an importance rank (0 = most important) and a direction (sign: 1 or -1). The rank swap flag is raised when the extracted top-k feature set differs from the top-k SHAP feature set. Internal ordering within that set is ignored, as defined in the previous section. A separate field, `unknown_features`, lists any feature-like names that do not match the valid feature set. A rule-based, deterministic comparison step checks each extracted field against ground-truth SHAP values and sets four binary flags. A narrative is marked as having at least 1 hallucination if at least one flag is set.

No human-coded subsample exists against which the pipeline can be calibrated, and prevalence rates should be read as automated-pipeline estimates. To quantify extraction reliability we adapt the semantic uncertainty approach of Kuhn et al. (Kuhn et al., 2023), submitting each extraction prompt five times at temperature 0.9 and measuring sign agreement and top-k set agreement across runs. The narrative reliability score is the arithmetic mean of these values, and narratives scoring below 0.8 are flagged as low-reliability and interpreted with caution.

4. Results

The pipeline generated 300 narratives (150 instances $\times$ 2 prompt strategies). The analysis covers the 272 narratives with valid extractions (130 direct, 142 chain-of-thought).¹ The results reveal that three-quarters of the valid narratives exhibit at least one type of hallucination. As shown in Table 2, rank swap emerged as the most frequent failure mode, followed by sign inversion and omission, while feature fabrication was entirely absent across both strategies. The 95% Wilson Confidence Interval is reported below in parentheses.

Extraction reliability, assessed on a 68-narrative subsample with five repeated runs at temperature 0.9, is high: mean narrative reliability score 0.946, with no narrative below the 0.8 threshold. Top- k set agreement is 0.903 for chain-of-thought versus 0.525 for zero-shot prompts. Because ambiguous narratives are harder to extract consistently, rank swap flags on narratives are a lower bound on true rank swap prevalence, and some true swaps will be missed. However, this

¹ Extraction failed for 9.3% of narratives (28/300), unevenly split across strategies (13.3% zero-shot vs 5.3% CoT). Five failures were HTTP 503 errors during the first minutes of evaluation (all zero-shot), likely reflecting endpoint instability rather than prompt strategy. Excluding those, the remaining failure rates (10.0% vs 5.3%) did not differ significantly ($\chi^2(1) \approx 1.9$, $p \approx .16$). We therefore treat extraction failures as missing data, not as evidence that one generation strategy produces systematically worse narratives.

conservative threshold guarantees that flagged swaps represent highly confident, substantive misalignments rather than artifacts of prose style or parser instability.

Table 3. Hallucination type rate by prompt strategy (Adult Income; Llama 3 70B; $n = 272$ valid extractions). Rows are ordered by overall prevalence. 95% Wilson confidence intervals in parentheses.

Type	Zero-shot ($n = 130$)	Chain-of-thought ($n = 142$)	Overall ($n = 272$)
Rank swap	80.8% (73.2-86.6%)	57.7% (49.5-65.6%)	68.8% (63.0-74.0%)
Sign inversion	60.0% (51.4-68.0%)	4.2% (2.0-8.9%)	30.9% (25.7-36.6%)
Omission	15.4% (10.2-22.6%)	19.0% (13.4-26.3%)	17.3% (13.3-22.2%)
Feature fabrication	0.0% (0.0-2.9%)	0.0% (0.0-2.6%)	0.0% (0.0-1.4%)
Any hallucination	93.1% (87.4-96.3%)	58.5% (50.2-66.2%)	75.0% (69.5-79.8%)

Because each Adult Income instance is processed under both strategies from the same SHAP table, the two columns of Table 3 are paired observations rather than independent samples. Strategy effects are therefore tested with McNemar's exact test on the 122 instances for which extraction succeeded under both strategies. Chain-of-thought reduces overall hallucination substantially: 40 instances are hallucinated under direct but not under chain-of-thought, while only 3 are hallucinated under chain-of-thought but not under direct. The reduction is driven almost entirely by sign inversion: 67 instances flip from hallucinated to faithful under chain-of-thought and none flip in the other direction. Rank swap also falls (33 instances improve under chain-of-thought versus 10 that worsen) but it remains the dominant failure mode under both strategies, with an absolute reduction of 23.1 percentage points. Two hallucination types show no significant difference between strategies: feature fabrication is zero under both, and omission rates are nearly identical at 15.4% versus 19.0%.

Table 4. McNemar exact tests comparing paired hallucination outcomes between prompt strategies (paired $n = 122$ instances with valid extractions under both strategies). b = zero-shot flagged, chain-of-thought not; c = chain-of-thought flagged, direct not. Risk difference is zero-shot – chain-of-thought on the unpaired Table 1 rates.

Hallucination type	b	c	$\chi^2(1)$	Exact two-sided p	Risk difference (pp)
Sign inversion	67	0	65.01	< .001	−55.8
Rank swap	33	10	11.26	< .001	−23.1
Feature fabrication	0	0	—	—	0.0
Omission	15	21	0.69	.405	+3.6
Any hallucination	40	3	30.14	< .001	−34.6

Sign inversion is a local, within-feature error: the model discusses the right feature but assigns it the wrong direction. Chain-of-thought’s explicit verification step, which requires the model to state the top feature and verify its sign before writing, drops this error from 60.0% to 4.2%. The intervention targets a specific, verifiable piece of information at generation time, and it works. Omission is strategy-invariant. Chain-of-thought’s ranking step identifies the top-five features by $|SHAP|$ but does not guarantee that all of them appear in the final narrative. The slight directional increase under chain-of-thought is consistent with token-length pressure as a contributing factor.

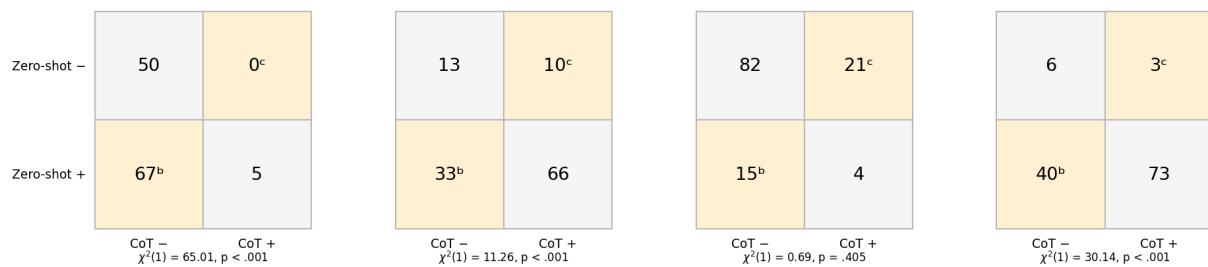


Figure 1. Paired McNemar contingency tables for each hallucination type ($n = 122$ paired instances with valid extractions under both prompt strategies).

Rank swap is a different kind of failure. It concerns which features the model treats as the most influential drivers of the prediction, not how it characterises any individual feature. Chain-of-thought reduces rank swap from 80.8% to 57.7%, but 57.7% remains a majority of narratives. When interpreting the empirical distribution of rank swap hallucinations across models, a fundamental

limitation must be considered: the recorded existence of these errors cannot entirely rule out foundational definitional ambiguity at its core. Because natural language summaries inherently balance structural fidelity with narrative synthesis, a recorded 'swap' might occasionally capture a divergence in stylistic prioritization rather than an objective factual breakdown. Further research is needed to analyze the semantic importance of 'rank' within open narratives, alongside how humans and autonomous systems interpret hierarchical weight. At this stage, rank swap metrics should be understood as capturing a combined spectrum of rigid sequence violations and fluid hierarchy re-alignments.

4.1. Sensitivity to k

The default $k = 3$ used in Tables 3 and 4 is one of several defensible choices. To assess whether the prompt-strategy effect on rank swap depends on this cutoff, we recomputed rank swap and omission rates at $k \in \{2, 3, 5\}$, holding all other pipeline elements fixed. Sign inversion and feature fabrication are k -invariant by definition and are not reported here. Rank swap and omission move in opposite directions as k grows (Figure 2). Enlarging the comparison set makes set-membership disagreement harder to trigger but coverage harder to satisfy, since the narrative must mention more of an 11-feature space. At $k = 5$, omission rises to 53.1% under zero-shot and 88.0% under chain-of-thought, while rank swap falls to 70.8% and 7.7% respectively. The two metrics are mechanically coupled at this boundary; reporting either in isolation at extreme k would mislead.

The prompt-strategy effect is itself k -sensitive. At $k = 2$, chain-of-thought offers no rank-swap advantage (75.4% vs 70.8%, overlapping CIs) and slightly elevates omission (14.1% vs 10.0%). At $k = 3$, chain-of-thought reduces rank swap by 23.1 percentage points with a 3.6-point omission cost. At $k = 5$, the rank-swap gap widens to 63.1 points but is mirrored by a 34.9-point omission gap in the opposite direction.

We report $k = 3$ as the default because it balances the two failure modes most evenly in this feature space. The rank-swap reduction we observe under chain-of-thought at $k = 3$ should not be assumed invariant to this choice. Studies on datasets with substantially different feature counts should re-tune k to their own feature space before comparing prevalence rates to this.

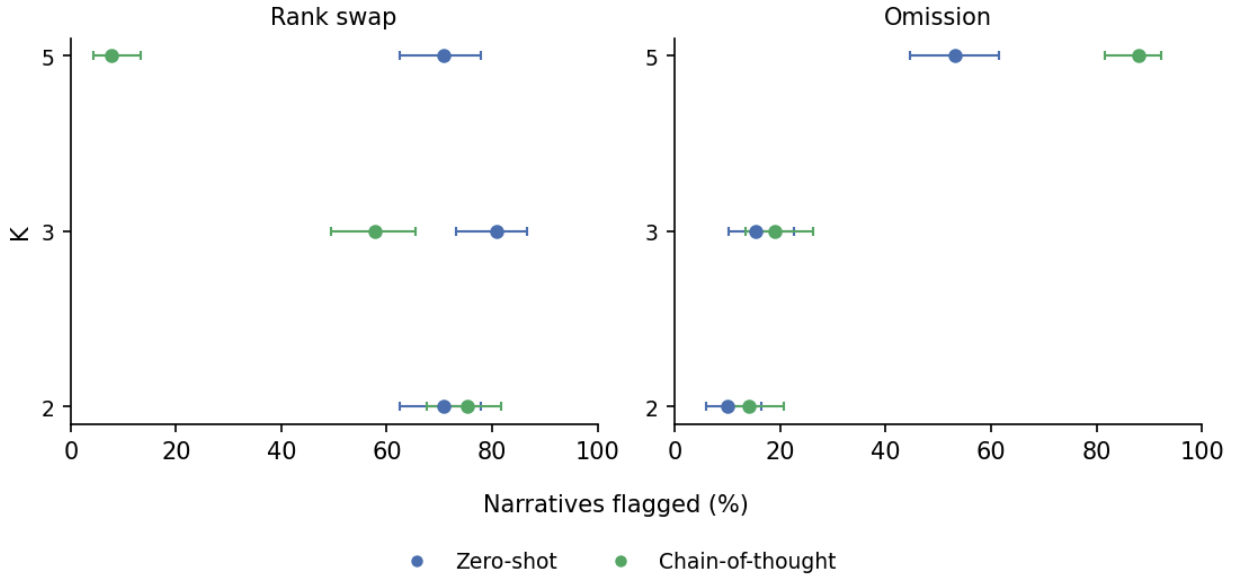


Figure 2. Rank swap and omission rates by k and prompt strategy, with 95% Wilson CIs ($n = 130$ zero-shot, $n = 142$ chain-of-thought).

4.2. Feature-level mention patterns

Regardless of the prompt strategy used, `hours_per_week` and `age` are nearly ubiquitous, appearing in 89.0% (242 mentions) and 83.5% (227 mentions) of narratives, respectively. In contrast, capital-related features are notably overlooked; `capital_gain` is included in only 19.5% of narratives and `capital_loss` in 22.1%, even when these features possess significant SHAP magnitudes. By prioritizing features with straightforward demographic associations, such as race, sex, or marital status, over capital indicators, narratives frequently fail to identify the true top- k drivers. This systematic preference directly inflates both omission rates and rank swap errors.

No narrative references a feature name outside the valid feature list, confirming the zero fabrication rate. The feature list in the prompt appears to function as a vocabulary constraint. No synonyms or paraphrases of feature names were detected in the `unknown_features` field.

The model appears to weight features by their interpretive strength (a person’s age and hours worked are easy to incorporate into a plausible income narrative) rather than by their SHAP magnitude. This is a hypothesis advanced by this paper rather than a directly tested finding. Isolating interpretive salience as a cause would require systematically varying which features carry large SHAP values while holding their narrative plausibility constant, and we leave this to follow-up work. Chain-of-thought’s ranking step partially corrects the bias by forcing the model to acknowledge which

features rank highest by SHAP before writing, but the final narrative still underrepresents capital features.

A second alternative hypothesis, which we cannot rule out, is corpus memorisation. The Adult Income dataset appears in many introductory machine-learning texts, tutorials, and blog posts and Llama-3-70B may be reproducing the rhetorical conventions of these discussions. Separating these two explanations requires either experimentation on datasets that post-date the model's training cutoff or are otherwise unlikely to appear in pretraining, or systematic variation of feature names while holding their statistical properties constant. Both are out of scope for this study.

5. Discussion and Conclusion

This paper provides a systematic measurement of faithfulness failure in LLM-generated SHAP narratives, revealing a substantial risk on this benchmark. On the Adult Income benchmark with Llama-3-70B, we observe high prevalence of three distinct error types under direct prompting: sign inversion (60.0%), rank swap (80.8%), and omission (15.4%). Chain-of-thought prompting reduces sign inversion to 4.2% (paired McNemar $p < .001$) but rank swap persists at 57.7%. Feature fabrication is absent under both conditions. We distinguish sign inversion and omission, which have unambiguous construct definitions, from rank swap, where set-membership disagreement may reflect either factual error or stylistic prioritisation. Although the rank-ser approach reduces subjectivity by clustering equivalent elements, rank swap errors still require nuanced interpretation because stylistic freedom clashes with factual fidelity. Future work should explore automated semantic weight mapping to isolate stylistic choices from structural errors and investigate how the perceived importance of rank varies across distinct narrative domains.

The finding that demographic features are systematically over-represented in narratives relative to capital-related features is a property of the prose we observed, not a property of user decisions. Whether readers of such narratives weight demographic attributes more heavily in their own decisions is an empirical question this study does not address. We flag it for user-study follow-up. The construct of 'explanatory discrimination', where a plausible but demographically-centred narrative masks the model's actual reasoning and induces unwarranted over-trust, would be the relevant frame for that follow-up work

We demonstrated that prompt engineering can effectively address specific failure modes, with chain-of-thought prompting nearly eliminating sign inversion. For practitioners, CoT is recommended for tasks requiring localized, verifiable accuracy, such as sign and direction. However, since rank swap persisted in 57.7% of CoT narratives, hallucination monitoring remains critical. We propose that the model appears to weight features by their interpretive salience rather than SHAP magnitude. This

interpretive salience remains an unconfirmed hypothesis that must be distinguished from the confirmed finding that capital features are quantitatively underrepresented in prose.

Finally, the observed zero fabrication rate requires an epistemic caveat. This finding may not represent a general property of LLM reliability but rather a vocabulary constraint imposed by the inclusion of the explicit feature list in the prompt. Future work should test whether removing this constraint or increasing feature-space complexity surfaces fabrication errors. Until then, the results suggest that while LLMs can be constrained to a valid vocabulary, ensuring they faithfully relay the structural truth of numeric grounding sources requires interventions beyond standard reasoning scaffolding.

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